

Solutions to Problem Set 1

1. For (a) and (b), we make use of the fact that

$$\sqrt{A} - B = (\sqrt{A} - B) \left(\frac{\sqrt{A} + B}{\sqrt{A} + B} \right) = \frac{A - B^2}{\sqrt{A} + B}.$$

(a)

$$\sqrt{x^2 + 1} - x = \frac{x^2 + 1 - x^2}{\sqrt{x^2 + 1} + x} = \frac{1}{\sqrt{x^2 + 1} + x}$$

$$\lim_{x \rightarrow \infty} (\sqrt{x^2 + 1} - x) = \lim_{x \rightarrow \infty} \frac{1}{\sqrt{x^2 + 1} + x} = 0.$$

(b) For $x > 0$, we have

$$\begin{aligned} \sqrt{x^2 + x} - x &= \frac{x^2 + x - x^2}{\sqrt{x^2 + x} + x} = \frac{x}{\sqrt{x^2 + x} + x} \\ &= \frac{1}{\sqrt{1 + \frac{1}{x}} + 1}. \end{aligned}$$

It follows that

$$\lim_{x \rightarrow \infty} (\sqrt{x^2 + x} - x) = \lim_{x \rightarrow \infty} \frac{1}{\sqrt{1 + \frac{1}{x^2}} + 1} = \frac{1}{1 + 1} = \frac{1}{2}.$$

(c) This is an indeterminate form of the type $0 \cdot \infty$. To evaluate the limit, we can rewrite it as a quotient and apply L'Hopital's rule. For $x > 0$, we have $\sqrt{x} \ln x = \frac{\ln x}{x^{-\frac{1}{2}}}$. We put $f(x) = \ln x$, $g(x) = x^{-\frac{1}{2}}$, and notice that

$$\lim_{x \rightarrow 0^+} f(x) = -\infty, \quad \lim_{x \rightarrow 0^+} g(x) = \infty.$$

We have

$$\lim_{x \rightarrow 0^+} \frac{f'(x)}{g'(x)} = \lim_{x \rightarrow 0^+} \frac{x^{-1}}{-\frac{1}{2}x^{-\frac{3}{2}}} = \lim_{x \rightarrow 0^+} \frac{\sqrt{x}}{-\frac{1}{2}} = 0.$$

It follows that

$$\lim_{x \rightarrow 0^+} \frac{f(x)}{g(x)} = 0,$$

by L'Hopital's rule.

- (d) This is again an indeterminate form of the type $0 \cdot \infty$, so we proceed as in part (c). We put

$$f(x) = \ln(x+5) - \ln x = \ln\left(1 + \frac{5}{x}\right), \quad g(x) = x^{-1},$$

and notice that

$$\lim_{x \rightarrow \infty} f(x) = \lim_{x \rightarrow \infty} g(x) = 0.$$

We have

$$\lim_{x \rightarrow \infty} \frac{f'(x)}{g'(x)} = \lim_{x \rightarrow \infty} \frac{\frac{-5x^{-2}}{(1+\frac{5}{x})}}{-x^{-2}} = \lim_{x \rightarrow \infty} \frac{5}{1 + \frac{5}{x}} = 5.$$

It follows that

$$\lim_{x \rightarrow \infty} \frac{f(x)}{g(x)} = 5,$$

by L'Hopital's rule.

- (e) Let $F(x) = (1 - \frac{2}{x} + \frac{7}{x^2})^x$. Then we have

$$\ln(F(x)) = x \ln\left(1 + \frac{3}{x} - \frac{5}{x^2}\right).$$

Using the approach of part (d), we find that

$$\lim_{x \rightarrow \infty} \ln(F(x)) = -2.$$

It follows that

$$\lim_{x \rightarrow \infty} F(x) = e^{-2}.$$

- (f) We put $f(x) = e^{3x} - 1 - 3x$, $g(x) = x^2$ and observe that

$$f'(x) = 3e^{3x} - 3, \quad f''(x) = 9e^{3x}, \quad g'(x) = 2x, \quad g''(x) = 2.$$

We have

$$\lim_{x \rightarrow 0} f(x) = \lim_{x \rightarrow 0} f'(x) = \lim_{x \rightarrow 0} g(x) = \lim_{x \rightarrow 0} g'(x) = 0.$$

Notice that

$$\lim_{x \rightarrow 0} \frac{f''(x)}{g''(x)} = \lim_{x \rightarrow 0} \frac{9e^{3x}}{2} = \frac{9}{2}.$$

Using L'Hopital's rule twice, we find that

$$\lim_{x \rightarrow 0} \frac{e^{3x} - 1 - 3x}{x^2} = \frac{9}{2}.$$

2. (a)

$$f'(x) = \frac{4x^3}{x^4 + 1}$$

(b)

$$f'(x) = 2x^3 \cos(2x) + 3x^2 \sin(2x) - 2xe^{x^2} \sin(e^{x^2})$$

(c)

$$f'(x) = \frac{(x^6 + 1)[-(\sqrt{x} + 3)e^{-x} + \frac{1}{2\sqrt{x}}e^{-x}] - 6x^5(\sqrt{x} + 3)e^{-x}}{(x^6 + 1)^2}$$

(d) We use logarithmic differentiation. Observe that

$$\ln(f(x)) = 3x \ln(1 + x^2).$$

Differentiating both sides we find that

$$\frac{f'(x)}{f(x)} = 3x \frac{2x}{1 + x^2} + 3 \ln(1 + x^2).$$

We conclude that

$$f'(x) = (1 + x^2)^{3x} \left[\frac{6x^2}{1 + x^2} + 3 \ln(1 + x^2) \right].$$

3. Observe that

$$F'(x) = f(x)g'(x) + f'(x)g(x),$$

$$F''(x) = f(x)g''(x) + f'(x)g'(x) + f'(x)g'(x) + f''(x)g(x) = f(x)g''(x) + 2f'(x)g'(x) + f''(x)g(x),$$

$$G'(x) = f'(g(x))g'(x),$$

$$G''(x) = f'(g(x))g''(x) + f''(g(x))g'(x)g'(x) = f'(g(x))g''(x) + f''(g(x))(g'(x))^2.$$

Substituting in the numbers, we find that

$$F''(1) = 3(-3) + 2(4)(-3) + 5(2) = -23,$$

$$G''(0) = 4(2) + 5(6)^2 = 188.$$

4. We begin by noting that

$$e^x = \sum_{k=0}^{\infty} \frac{x^k}{k!}.$$

It follows that

$$e^{-x^5} = \sum_{k=0}^{\infty} \frac{(-1)^k x^{5k}}{k!},$$

and consequently

$$f(x) = x^3 \sum_{k=0}^{\infty} \frac{(-1)^k x^{5k}}{k!} = \sum_{k=0}^{\infty} \frac{(-1)^k x^{3+5k}}{k!}.$$

The coefficient of x^{43} in the above series for $f(x)$ is $\frac{1}{8!}$. (Notice that x^{43} corresponds to $k = 8$.) Since this coefficient must also be equal to

$$\frac{f^{(43)}(0)}{43!},$$

we conclude that

$$f^{(43)}(0) = \frac{43!}{8!}.$$

5. Let x be the distance from point to where the banker lands, and let us denote the distance that the banker swims by z . By the Pythagorean theorem,

$$z = \sqrt{x^2 + 1000^2}.$$

The distance that the banker runs is $2000 - x$. The total time that it takes to reach the client is therefore given by

$$T(x) = \frac{\sqrt{x^2 + 1000^2}}{3} + \frac{2000 - x}{5}.$$

Taking the derivative of T , we obtain

$$T'(x) = \frac{x}{3\sqrt{x^2 + 1000^2}} - \frac{1}{5}.$$

Setting $T'(x) = 0$, we find that

$$\frac{x}{\sqrt{x^2 + 1000^2}} = \frac{3}{5}.$$

Solving for x we find that

$$\frac{x^2}{x^2 + 1000^2} = \frac{9}{25},$$

or

$$x^2 \left(\frac{16}{25} \right) = \left(\frac{9}{25} \right) 1000^2.$$

Since $x > 0$, we conclude that

$$x = \left(\frac{3}{4} \right) 1000 = 750.$$

One can verify that this indeed provides a minimum by checking the second derivative.

6. (a) Observe that

$$\begin{aligned} f'(x) &= \frac{1}{x+1} = (x+1)^{-1}, \\ f''(x) &= -(x+1)^{-2}, \end{aligned}$$

$$f'''(x) = 2(x+1)^{-3},$$

$$f^{(4)}(x) = -3 \cdot 2(x+1)^{-4},$$

and, in general,

$$f^{(n)}(x) = (-1)^{n-1}(n-1)!(x+1)^{-n},$$

so that

$$f^{(n)}(0) = (-1)^{n-1}(n-1)!, \quad n = 1, 2, 3, \dots$$

Observe also that $f(0) = \ln 1 = 0$. We conclude that the Taylor series for f about 0 is given by

$$f(x) = \sum_{k=1}^{\infty} (-1)^{k-1} \frac{x^k}{k}.$$

(b) By substitution, we find that the Taylor series for f is given by

$$f(x) = \sum_{k=1}^{\infty} (-1)^{k-1} \frac{x^{2k}}{k}.$$

7. (a)

$$\int (\sqrt{x} + 1)^2 dx = \int (x + 2\sqrt{x} + 1) dx = \frac{1}{2}x^2 + \frac{4}{3}x^{\frac{3}{2}} + x + C$$

(b) Let $z = \sqrt{x}$, so that $x = z^2$ and $dx = 2zdz$. Then we have

$$\int e^{\sqrt{x}} dx = 2 \int ze^z dz.$$

The integral on the right can be computed using integration by parts. Let $u = z$ and $dv = e^z dz$ so that

$$\begin{aligned} \int ze^z dz &= ze^z - \int e^z dz \\ &= ze^z - e^z + C. \end{aligned}$$

We conclude that

$$\begin{aligned} \int e^{\sqrt{x}} dx &= 2ze^z - 2e^z + C \\ &= 2\sqrt{x}e^{\sqrt{x}} - 2e^{\sqrt{x}} + C. \end{aligned}$$

Comment: This integral can also be evaluated by parts (without first making a substitution) by putting

$$u = \sqrt{x}, \quad dv = \frac{e^{\sqrt{x}}}{\sqrt{x}} dx.$$

(c) We use integration by parts twice. Let

$$u = (\ln x)^2, \quad dv = x^4 dx,$$

so that

$$du = \frac{2}{x} \ln x dx, \quad v = \frac{1}{5} x^5.$$

Then we have

$$\int x^4 (\ln x)^2 dx = \frac{1}{5} x^5 (\ln x)^2 - \frac{2}{5} \int x^4 \ln x dx.$$

Now let

$$U = \ln x, \quad dV = x^4 dx,$$

so that

$$dU = \frac{dx}{x}, \quad V = \frac{1}{5} x^5.$$

Then we have

$$\begin{aligned} \int x^4 \ln x dx &= \frac{1}{5} x^5 \ln x - \frac{1}{5} \int x^4 dx \\ &= \frac{1}{5} x^5 \ln x - \frac{1}{25} x^5 + C. \end{aligned}$$

It follows that

$$\int x^4 (\ln x)^2 dx = \frac{1}{5} x^5 (\ln x)^2 - \frac{2}{25} x^5 \ln x + \frac{2}{125} x^5 + C.$$

(d) Let $u = x + 1$ so that $x = u - 1$ and $dx = du$. Then we have

$$\begin{aligned} \int \frac{x^2}{\sqrt{x+1}} dx &= \int \frac{(u-1)^2}{\sqrt{u}} du = \frac{u^2 - 2u + 1}{\sqrt{u}} du \\ &= \left(\int u^{\frac{3}{2}} - 2u^{\frac{1}{2}} + u^{-\frac{1}{2}} \right) du \\ &= \frac{2}{5} u^{\frac{5}{2}} - \frac{4}{3} u^{\frac{3}{2}} + 2u^{\frac{1}{2}} + C \\ &= \frac{2}{5} (x+1)^{\frac{5}{2}} - \frac{4}{3} (x+1)^{\frac{3}{2}} + 2(x+1)^{\frac{1}{2}} + C. \end{aligned}$$

(e) We use integration by parts. Let

$$u = x^2, \quad dv = x \cos(x^2),$$

so that

$$du = 2x dx, \quad v = \frac{1}{2} \sin(x^2).$$

Then we have

$$\begin{aligned}\int x^3 \cos(x^2) dx &= \frac{1}{2} x^2 \sin(x^2) - \int x \sin(x^2) dx \\ &= \frac{1}{2} x^2 \sin(x^2) + \frac{1}{2} \cos(x^2) + C.\end{aligned}$$

8. (a) We first compute the indefinite integral. We let $u = x + 1$ so that $x = u - 1$ and $dx = du$. Then we have

$$\begin{aligned}\int \frac{x}{\sqrt{x+1}} dx &= \int \frac{u-1}{\sqrt{u}} du = \int (u^{\frac{1}{2}} - u^{-\frac{1}{2}}) du \\ &= \frac{2}{3} u^{\frac{3}{2}} - 2u^{\frac{1}{2}} + C \\ &= \frac{2}{3} (x+1)^{\frac{3}{2}} - 2(x+1)^{\frac{1}{2}} + C\end{aligned}$$

It follows that

$$\begin{aligned}\int_1^3 \frac{x}{\sqrt{x+1}} dx &= \left(\frac{2}{3} (x+1)^{\frac{3}{2}} - 2(x+1)^{\frac{1}{2}} \right) \Big|_1^3 \\ &= \frac{2}{3} (4^{\frac{3}{2}} - 2^{\frac{3}{2}}) - 2(\sqrt{4} - \sqrt{2}) \\ &= \frac{2}{3} (8 - 2^{\frac{3}{2}}) - 2(2 - \sqrt{2}).\end{aligned}$$

- (b) Observe that

$$\int_{-1}^1 \sqrt{x^2 + x^4} dx = 2 \int_0^1 \sqrt{x^2 + x^4} dx.$$

For $x \geq 0$, we have

$$\sqrt{x^2 + x^4} = \sqrt{x^2(1 + x^2)} = x\sqrt{1 + x^2},$$

so that

$$\int \sqrt{x^2 + x^4} dx = \int x\sqrt{1 + x^2} dx = \frac{1}{3} (1 + x^2)^{\frac{3}{2}} + C.$$

We conclude that

$$\int_{-1}^1 \sqrt{x^2 + x^4} dx = \frac{2}{3} (1 + x^2)^{\frac{3}{2}} \Big|_0^1 = \frac{2}{3} (2^{\frac{3}{2}} - 1)$$

(c) Observe that

$$\frac{e^x + 5}{e^{4x}} = e^{-3x} + 5e^{-4x},$$

so that

$$\begin{aligned}\int_0^{\ln 2} \frac{e^x + 5}{e^{4x}} dx &= \int_0^{\ln 2} (e^{-3x} + 5e^{-4x}) dx \\&= \left(-\frac{1}{3}e^{-3x} - \frac{5}{4}e^{-4x} \right) \Big|_0^{\ln 2} \\&= -\frac{1}{3}e^{-3\ln 2} - \frac{5}{4}e^{-4\ln 2} + \frac{1}{3} + \frac{5}{4} \\&= \frac{7}{24} + \frac{75}{64}.\end{aligned}$$

9. Observe that

$$\int_1^2 f(x) dx = \int_0^2 f(x) dx - \int_0^1 f(x) dx = 3 - 5 = -2,$$

and consequently

$$\int_1^3 f(x) dx = \int_1^2 f(x) dx + \int_2^3 f(x) dx = -2 + 5 = 3.$$

10. Using integration by parts, we find that

$$\begin{aligned}\int_0^1 f(x) dx &= x f(x) \Big|_0^1 - \int_0^1 x f'(x) dx \\&= f(1) - 0 - \int_0^1 x e^{-x^2} dx \\&= \frac{1}{2} e^{-x^2} \Big|_0^1 = \frac{1}{2}(e^{-1} - 1).\end{aligned}$$

11. (a)

$$G'(x) = -\frac{d}{dx} \int_5^x e^{t^3} dt = -e^{x^3}.$$

(b)

$$\begin{aligned}G'(x) &= \sqrt{1 + (\sqrt{x})^2} \frac{d}{dx} \sqrt{x} - \sqrt{1 + (\ln x)^2} \frac{d}{dx} \ln x \\&= \sqrt{1+x} \left(\frac{1}{2\sqrt{x}} \right) - \sqrt{1 + (\ln x)^2} \left(\frac{1}{x} \right).\end{aligned}$$

12. Let f be a continuous function such that

$$n \int_x^{x+\frac{1}{n}} f(t) dt = f(x) + \frac{1}{n} \quad (1)$$

for every $x \in \mathbb{R}$ and every positive integer n . By the fundamental theorem of calculus, we conclude from (1) that f is differentiable and

$$n[f(x + \frac{1}{n}) - f(x)] = f'(x) \quad (2)$$

for every $x \in \mathbb{R}$ and every positive integer n . Using (2) and the fact that f is differentiable, we conclude that f is twice differentiable and that

$$n[f'(x + \frac{1}{n}) - f'(x)] = f''(x) \quad (3)$$

for every $x \in \mathbb{R}$ and every positive integer n . It follows from (3) that f'' is continuous.

Applying Taylor's Theorem with $n = 1$ to (2) we conclude that for every $x \in \mathbb{R}$ and every positive integer n , we may choose a point

$$c_{x,n} \in (x, x + \frac{1}{n})$$

such that

$$n[f(x) + f'(x)\frac{1}{n} + \frac{1}{2n^2}f''(c_{x,n}) - f(x)] = f'(x).$$

It follows that $f''(c_{x,n}) = 0$ for every $x \in \mathbb{R}$ and every positive integer n . Since f'' is continuous, we conclude that $f''(x) = 0$ for all $x \in \mathbb{R}$ and consequently there are constants a and b such that

$$f(x) = ax + b \quad \text{for all } x \in \mathbb{R}.$$

(Notice that $c_{x,n} \rightarrow x$ as $n \rightarrow \infty$ and consequently $f''(c_{x,n}) \rightarrow f''(x)$ as $n \rightarrow \infty$.)

Substituting $f(x) = ax + b$ into (1) we find that

$$n[\frac{a}{2}(x + \frac{1}{n})^2 - \frac{a}{2}x^2 + b(x + \frac{1}{n}) - bx] = ax + b + \frac{1}{n}.$$

We conclude that $a = 2$. Thus a continuous function f satisfies (1) if and only if there is a constant b such that $f(x) = 2x + b$ for all $x \in \mathbb{R}$.